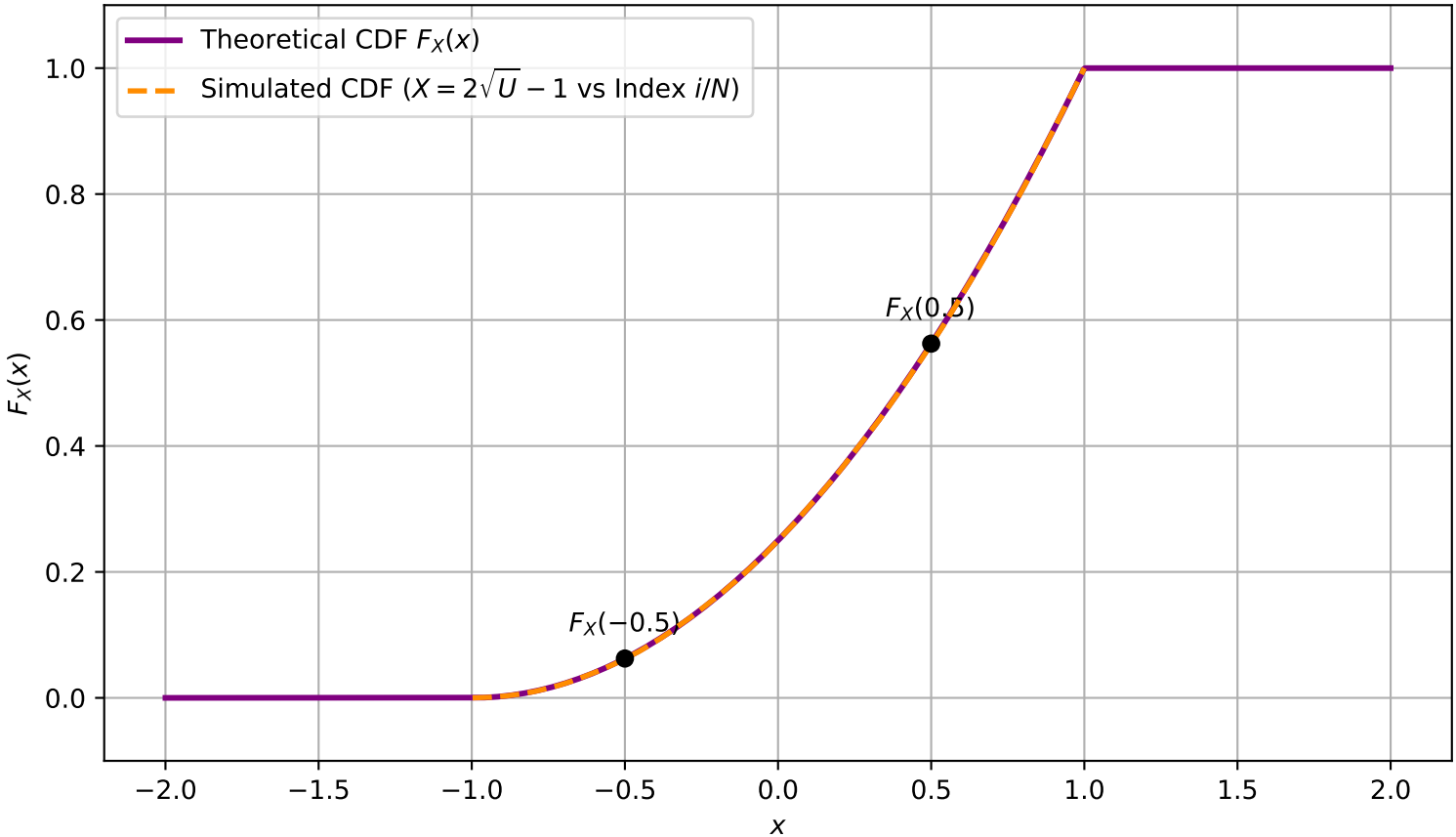
CDF: $\Pr(X^2 \le 0.25) = F_X(0.5) - F_X(-0.5) = 0.5$
Theoretical CDF $F_X(x)$
Simulated CDF ($X = 2\sqrt{U} - 1$ vs Index $i/N$)
$F_X(0.5)$
$F_X(-0.5)$
$F_X(x)$
$x$
1.0
0.8
0.6
0.4
0.2
0.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0